

SUJAL PATIL

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ABOUT

Computer Science student skilled in Java, Spring Boot, PostgreSQL, REST APIs, and WebSockets, building scalable real-time and database-driven applications.

EDUCATION

Dr D.Y Patil Dnyaan Prasad Global University, Pimpri Chinchwad (2025 – 2028)

B.Tech, Computer Science & Engineering | CGPA: 8.73

Saraswati Institute of Technology, Kharghar (2022 – 2025)

Diploma in Computer Engineering | 92.63%

SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript
- **Backend:** Spring Boot, Hibernate, REST APIs, WebSockets
- **Frontend:** React, HTML, CSS, Responsive Web Design
- **Databases:** PostgreSQL, MySQL
- **Machine Learning:** XGBoost, DBSCAN, Clustering
- **Tools & Platforms:** Git, GitHub, Maven, Postman, Railway, Cloudflare Pages, k6

EXPERIENCE

Web Developer Intern — Saliency Foundation

March 2026 – June 2026

- Contributed to the development of the company's new website by building responsive web pages and integrating REST APIs to deliver dynamic content across 10+ pages.
- Diagnosed and resolved frontend-backend integration issues and REST API communication bugs, improving application stability and ensuring consistent functionality across devices.

PROJECTS

Drill Insight — Oil Well Intelligence System / *React, FastAPI, PostgreSQL, XGBoost* / [GitHub](#)

- Analyzed **203 wells and 3,300+ drilling events** using PostgreSQL and Python-based data processing to identify historical drilling patterns, nearby-well relationships, and operational risks.
- Developed an **XGBoost-based recommendation model** combining geological, reservoir, historical drilling, and offset-well features to support data-driven new-well location decisions.
- Built data-driven analytical views for **historical trends, well comparisons, geospatial relationships, and operational risk assessment** to support drilling decision-making.

GeoWatch / *Spring Boot, PostgreSQL, WebSockets, DBSCAN, Leaflet* / [GitHub](#) , [Demo](#)

- Built a real-time crowd safety platform that identifies high-risk zones during public events by analyzing live GPS incident reports and automatically generating geospatial risk clusters for organizers.
- Implemented **DBSCAN-based** hotspot detection with a custom grid-based spatial index, reducing neighbor-search overhead and achieving **0.92 ms** clustering execution under benchmark load.
- Designed a scalable event processing and WebSocket broadcasting pipeline supporting **500+ concurrent clients**; benchmarked using k6, processing **88,000+ REST requests** at **732.99 req/sec** under **250 users**.

GitGo / *TypeScript, VS Code Extension API* / [GitHub](#) [Marketplace](#)

- Developed a **VS Code Extension** that automates coding problem publication workflows, consolidating folder creation, README generation, screenshot management, and Git operations into a **single-command pipeline**.
- Designed a modular transactional architecture with **automatic rollback** and repository analytics, enabling scans of **10,000+ problem folders** in under 2.3 seconds.

EXTRACURRICULAR ACTIVITIES

- Solved **200+ problems** on [LeetCode](#).
- Technical Team Member at Kryptonex Tech & Innovation Club, organizing tech events for 100+ participants.
- Received the Best Website award at WebCraft Carnival 2026.
- Co-founded [RuntimeSync](#), a GitHub organization for collaborative development and repository management.

CERTIFICATIONS

- Oracle Cloud Infrastructure Foundations Associate (2025) [Credential](#)
- Google Agentic AI — Google x Kaggle (2025) [Credential](#)